

Bloom

A bracelet that blooms when two
people truly meet

Alex Tully, Terra Labs, USC, Fall 2026



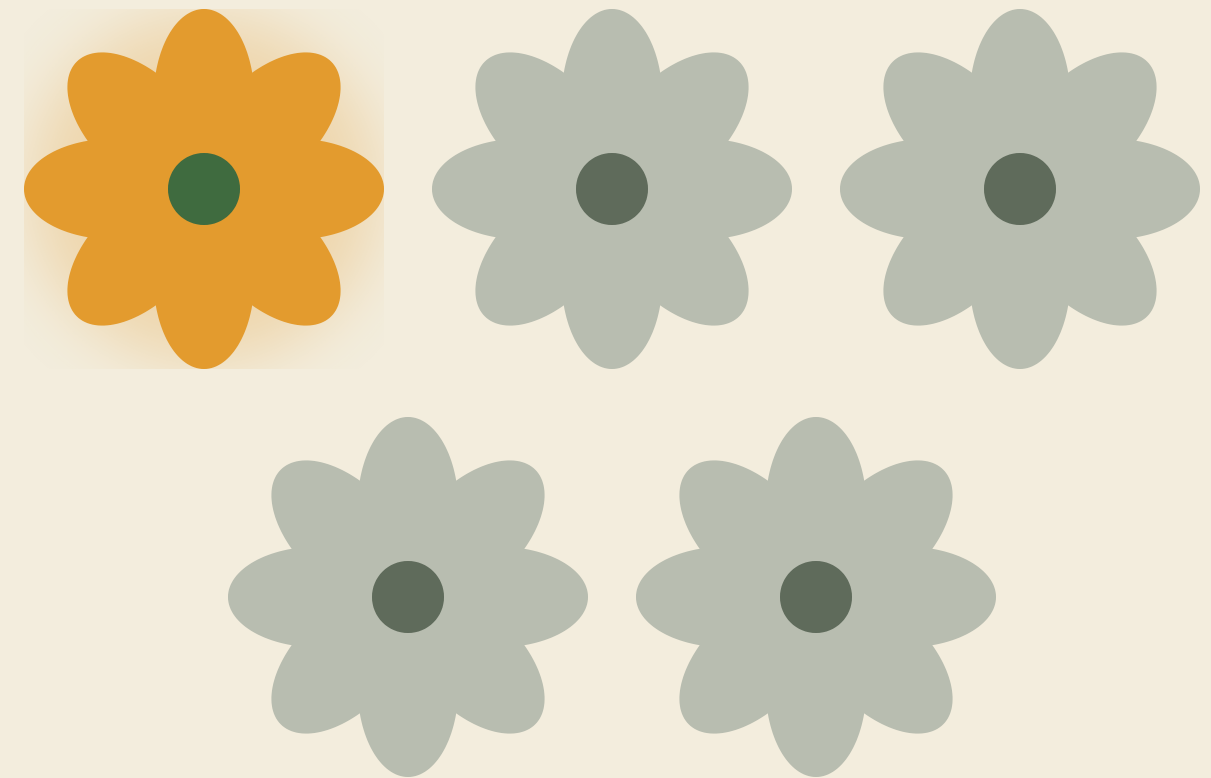
You meet more people than you can keep

80%

of new professional contacts go dormant

The context that made them valuable is lost within 48 hours

20 to 40 new connections a month for students, founders and community builders



Terra's Garden worked, except for one step

Built in a 5 hour hackathon

Check in

Guest creates a profile and is planted as a seed

Meet someone

They tap your bracelet's NFC tag with their phone

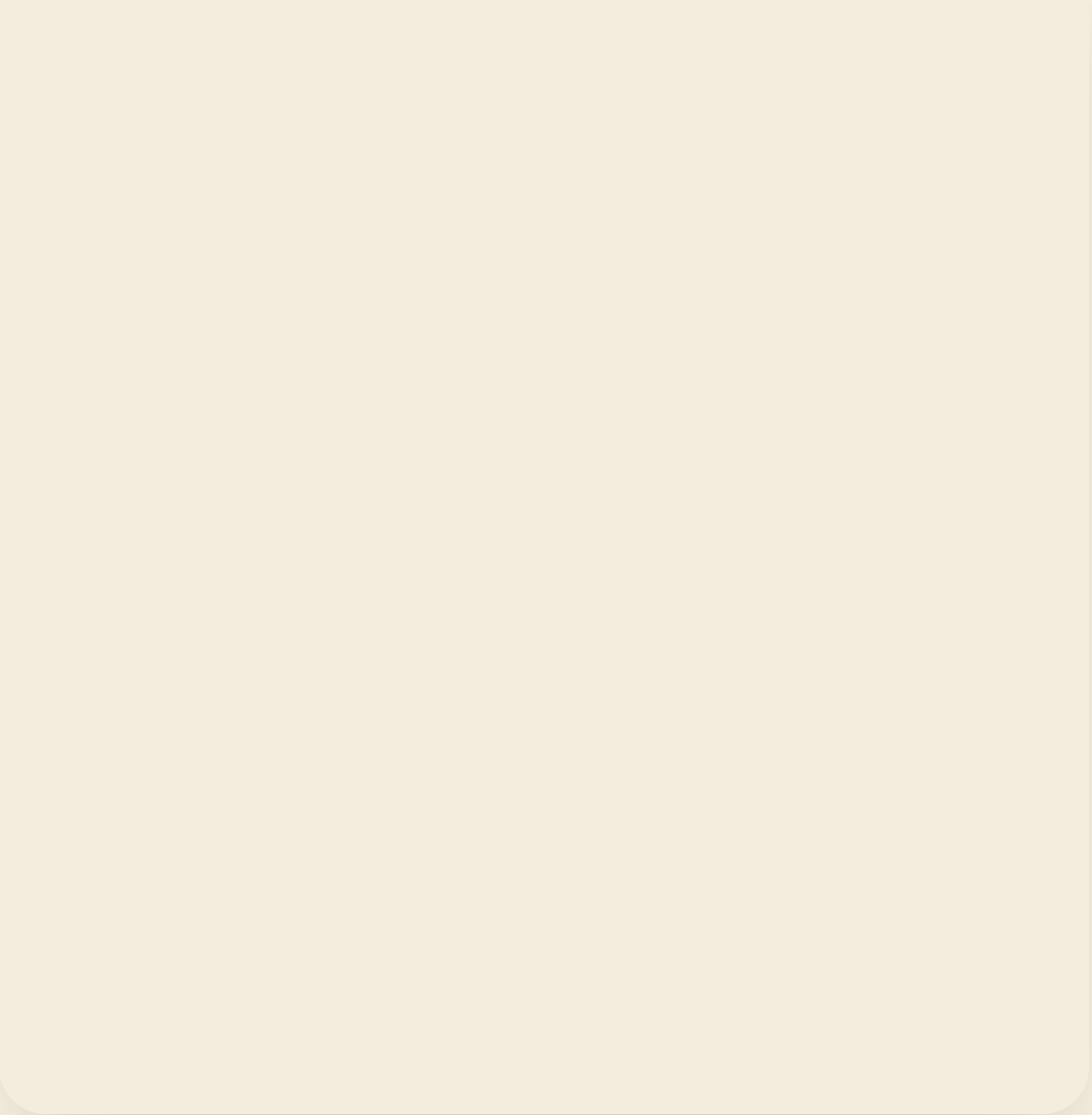
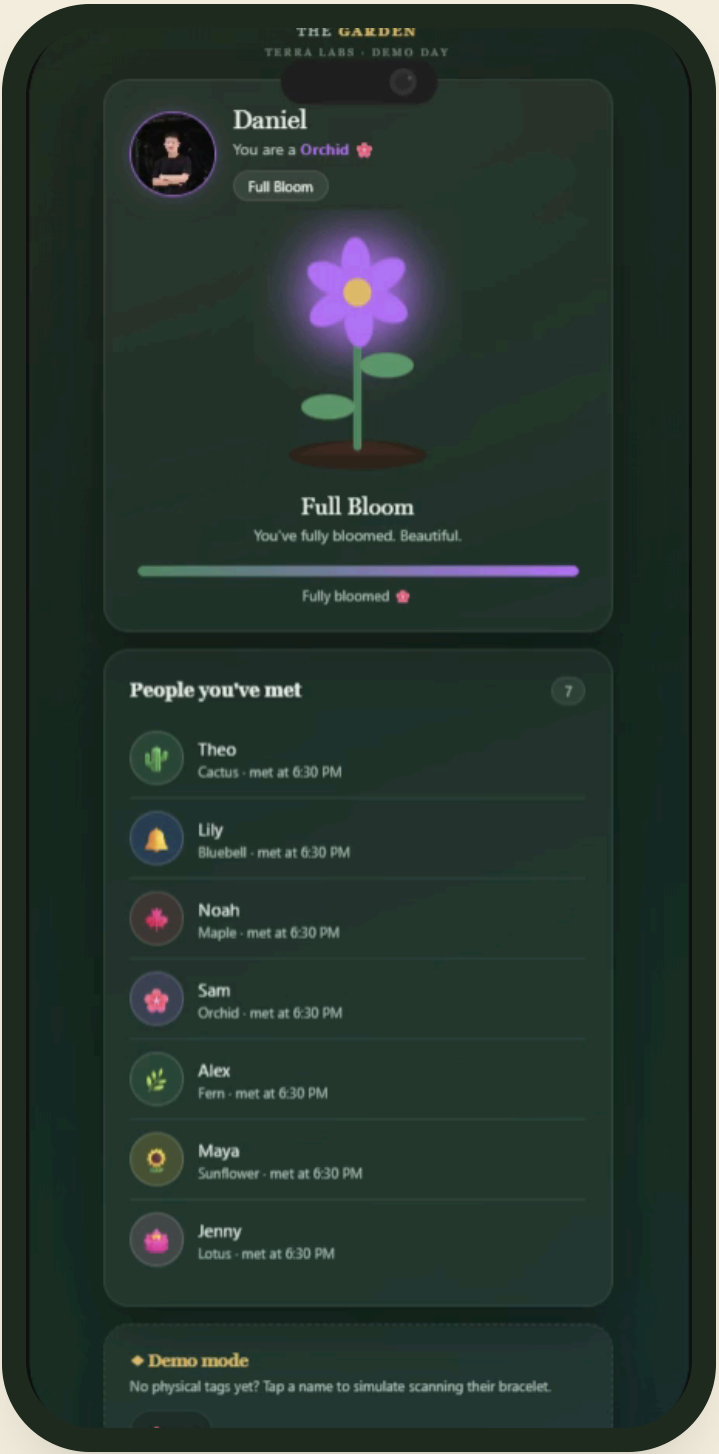
Grow

Your bracelet glows brighter; 7 connections reach full bloom

Garden

Every flower joins a projected communal garden

Pulling out a phone breaks the moment



The bracelet listens for one thing

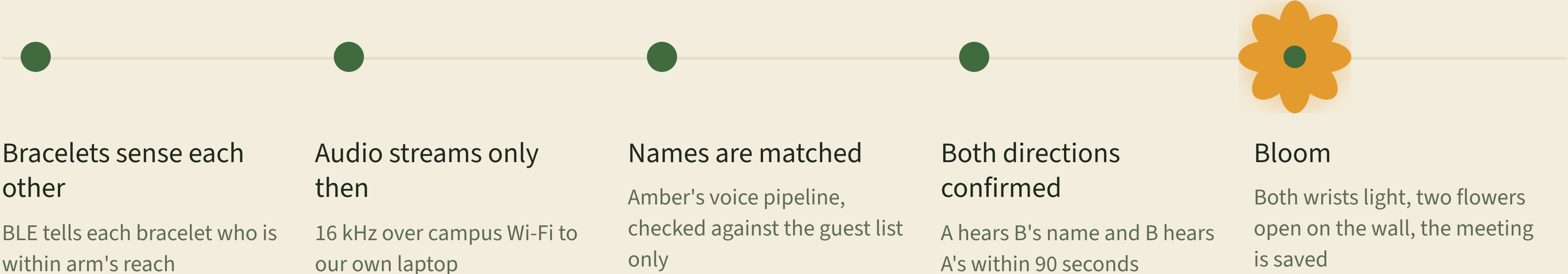
Two people tell each other their names.

Only then do both bracelets bloom.

Every person you bloomed with is saved to the Garden site for the next morning.



How a bloom happens



ESP32-S3 joins WPA2-Enterprise campus Wi-Fi; phone hotspots are the fallback

A device that listens has to earn it



Consent at check-in

You choose to wear it, and you can opt out of saving anything



A visible indicator

The petals breathe while the microphone is live



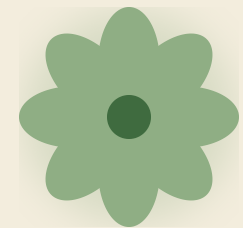
Hold to mute

Petals turn amber and the microphone is off



No audio is ever stored

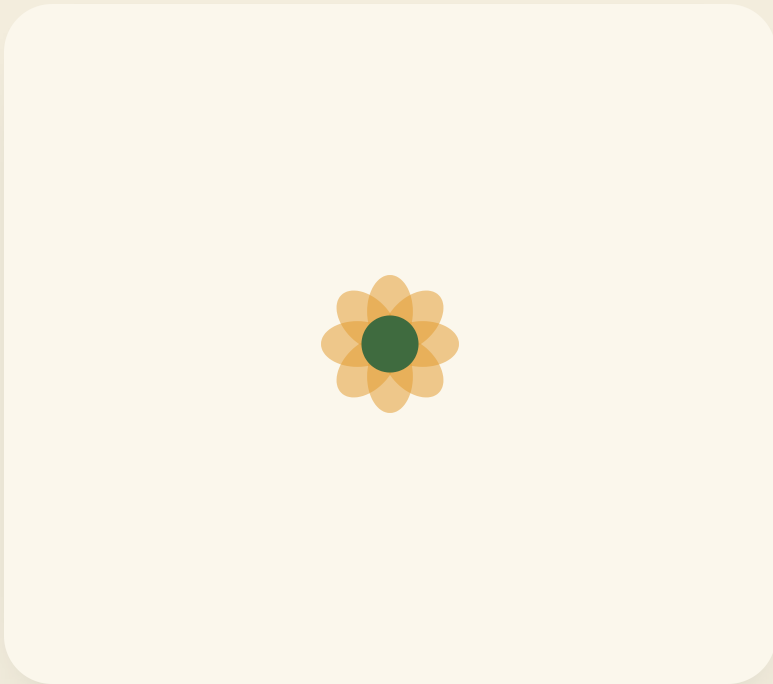
Only 'Alex met Maya at 7:12' is written down



A two-person fallback

Both press the button within 3 seconds and it blooms anyway

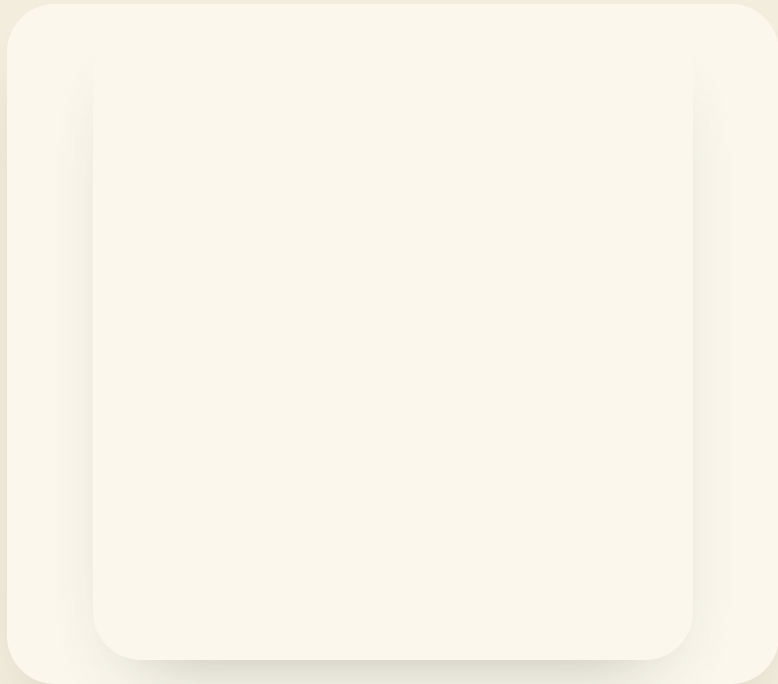
One guest's night



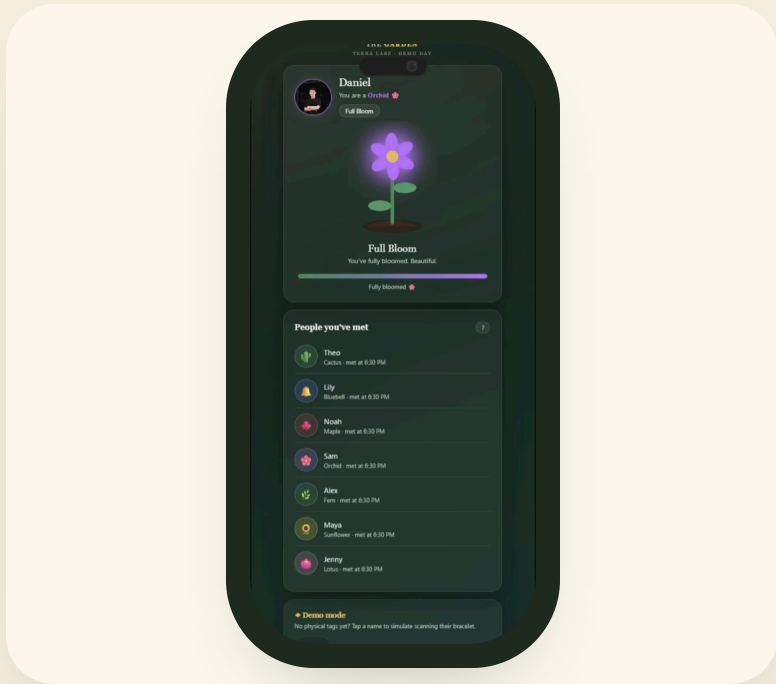
7:00
Planted
Name, photo, one fun fact. A dark seed goes on your wrist.



7:12
First bloom
You and Maya swap names. Both wrists open, two flowers unfurl on the wall.



8:30
The garden fills
The wall shows what the room has become.



Next morning
Everyone you met
The Garden site lists each person, with their face and fun fact.

We are not building the brains

Amber is live

A relationship memory app in TestFlight that turns speech into a people graph

It has an API

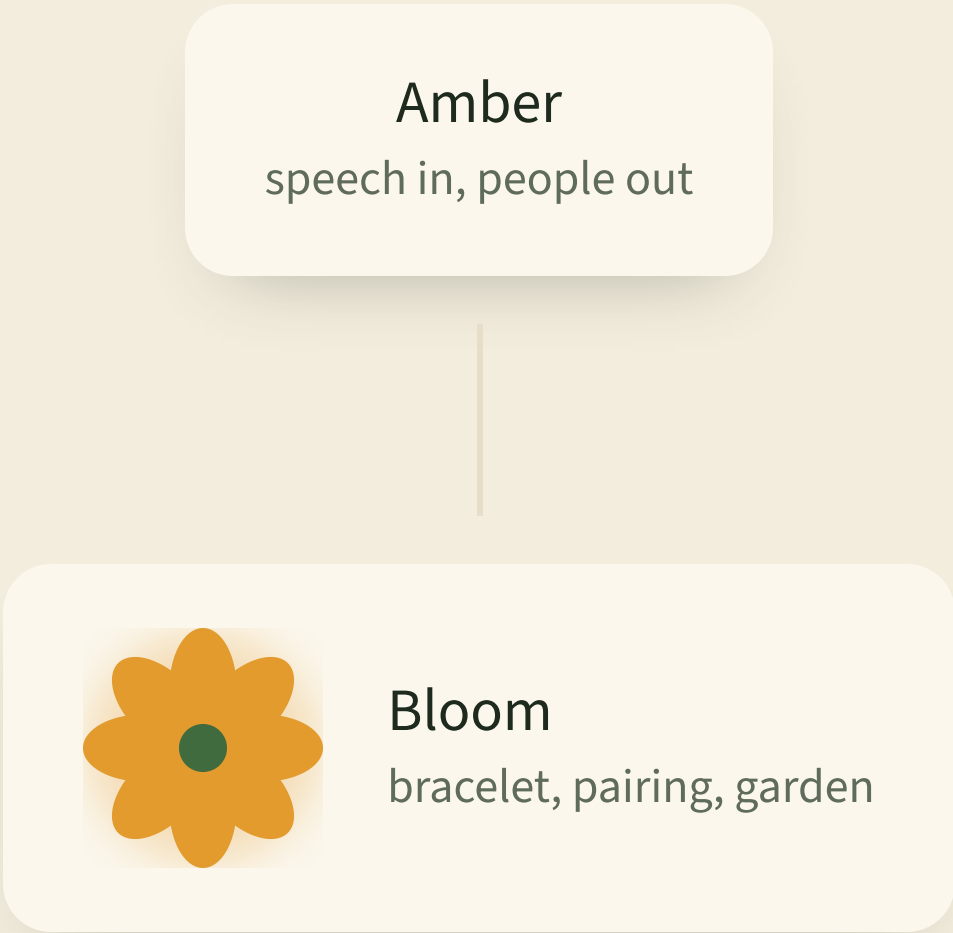
We call the pipeline that already exists instead of writing one

Its wearable is on the roadmap

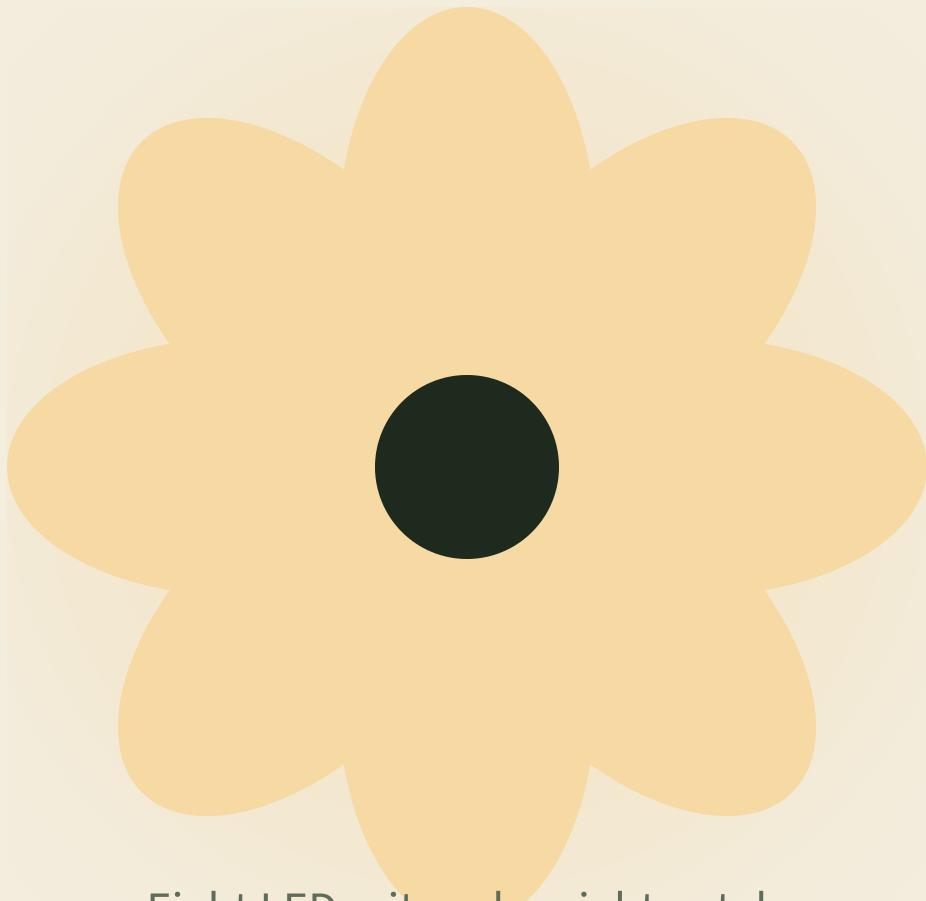
First prototype targeted December 2026, the month of our demo

It is a partner, not a dependency

Every Amber call has a Deepgram and own-database fallback



What is on the wrist



Eight LEDs sit under eight petals

- XIAO ESP32-S3 \$7.49
Wi-Fi, Bluetooth, I2S audio, battery charger
- 8 addressable LED petals
WS2812B, the same family as the hackathon
- One button
Hold to mute, press together to handshake

- I2S MEMS microphone
Digital, omnidirectional
- 500 or 800 mAh LiPo \$7.16
3 to 5 hours with proximity-gated streaming
- Printed PETG body and petals
Elastic strap, under 40 g

Lean

\$944

32 bracelets, about \$30 each all in
Off-the-shelf boards, hand-wired

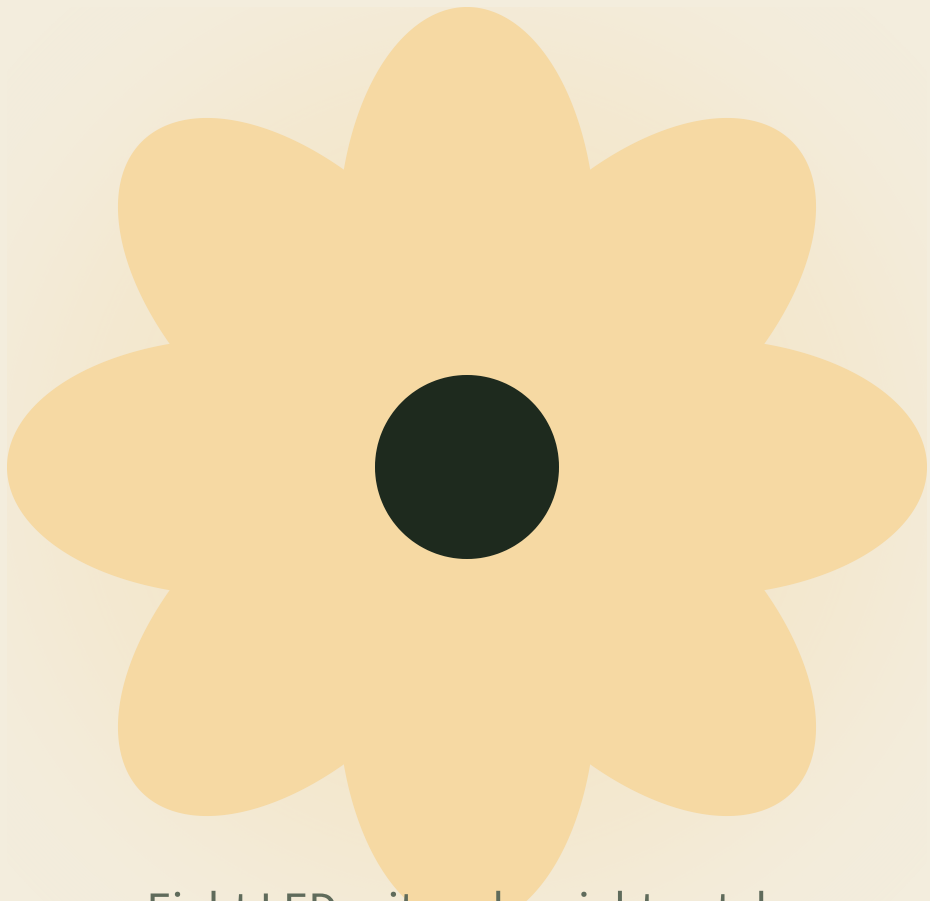
Mid

\$2,197

66 bracelets, about \$33 each all in
Custom carrier PCB assembled by JLCPCB,
plus haptics

No router and no extra PC: campus Wi-Fi and our own laptop

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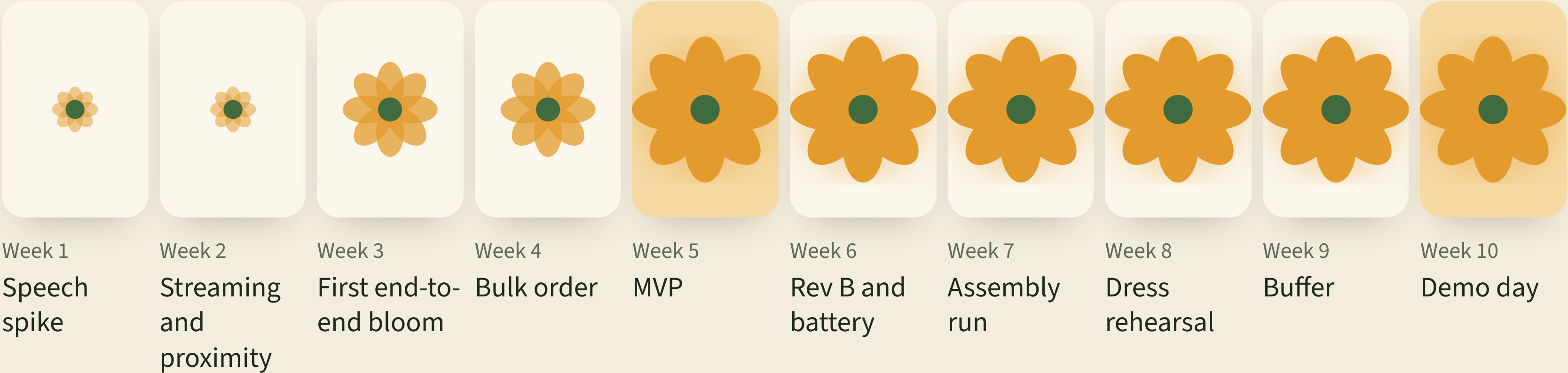
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Sources: Bloom BOM F26, team estimates, prices checked Sep 22 2026; seeedstudio.com, checked Sep 22 2026; adafruit.com, checked Sep 22 2026; deepgram.com/pricing, checked Sep 22 2026

10 weeks to demo day

MVP Fri Oct 30, 2026. Demo Fri Dec 4, 2026.



Who we need



PM and systems
Alex Tully



Firmware
open seat



Web and projection
open seat



Narrative and visual design
open seat



Hardware and electrical
open seat



Backend and Amber
integration
open seat



Industrial design
open seat

8.5

hours a week, per person

6 to 10 hours a week, 12 to 15 in the MVP,
assembly and demo weeks

About 564 team hours across the semester

What could go wrong

Names are missed in a loud room	Week 1 spike, roster-constrained matching, button fallback
Three people in a circle confuse it	Proximity ranking and both directions required; tuned at rehearsal
Campus Wi-Fi blocks 60 devices	Tested in the demo room in week 7, phone hotspots ready
Amber is not available when we need it	Deepgram plus our own database works from week 3
Our laptop dies on the night	Second laptop with the same image; the button works with no server

What we are asking for

A team

Five to seven people, about 8 hours a week

A budget tier

Lean at \$944 or Mid at \$2,197

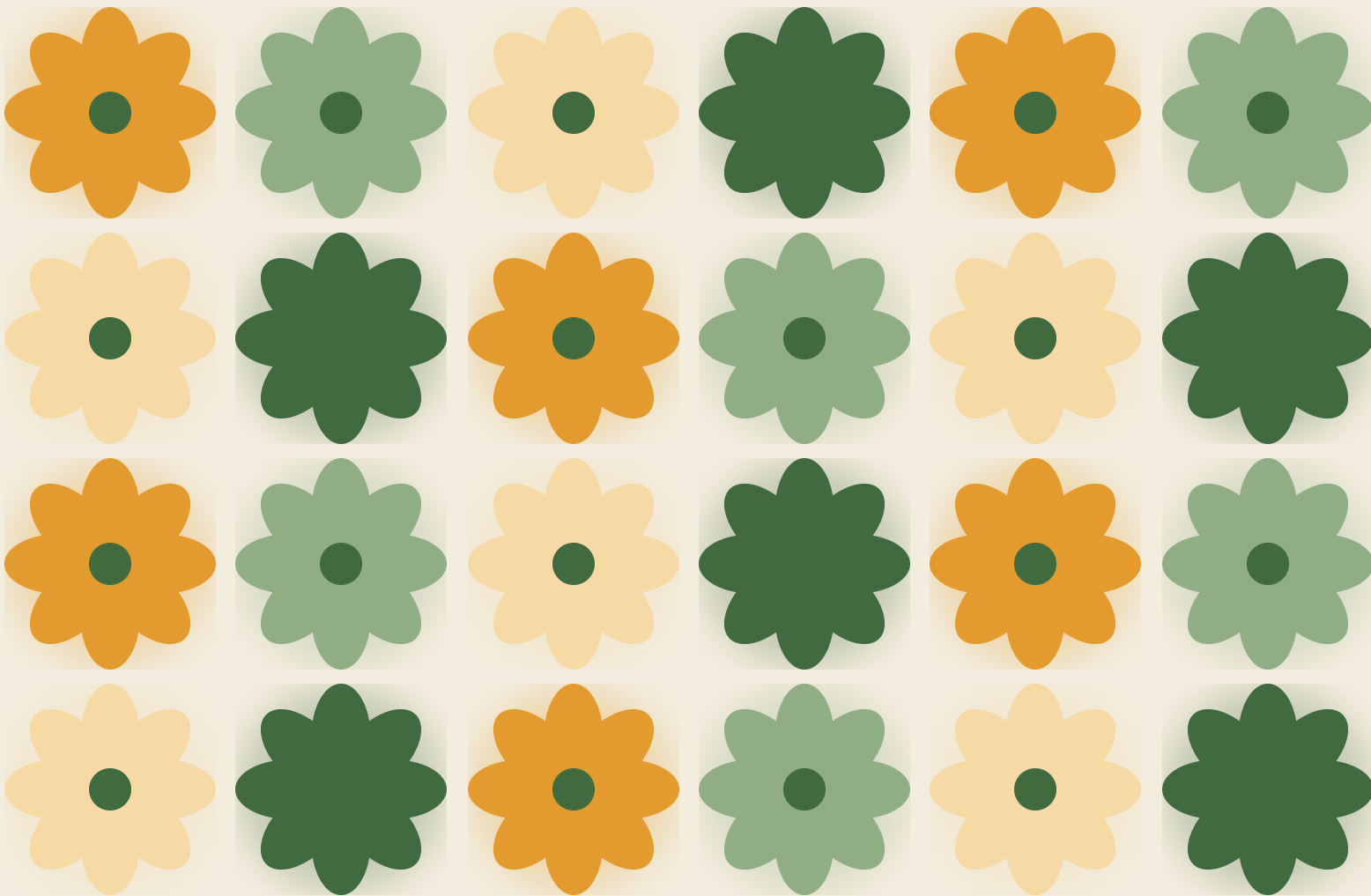
A narrative partner

To make the forest listen kindly

A wall

One projection surface and a check-in table by the door

Let the forest bloom when people meet.



Sources

Amber Intelligence deck, 2026

Amber Intelligence investor deck, slides 'Your network has a churn problem' and 'Our Product'

Bloom BOM F26, team estimates, prices checked Sep 22 2026

Bloom bill of materials, Lean and Mid tiers, Terra Labs template

adafruit.com, checked Sep 22 2026

Adafruit Lithium Ion Polymer Battery 3.7 V 500 mAh, product 1578

<https://www.adafruit.com/product/1578>

Bloom Task Sheet F26, team plan

Bloom week-by-week task sheet, Terra Labs template, 70 tasks

Terra's Garden deck, Team Steve, Sep 2026

The Terra-rium: Terra's Garden hackathon presentation (5 hour build)

seeedstudio.com, checked Sep 22 2026

Seeed Studio XIAO ESP32-S3 product page

<https://www.seeedstudio.com/XIAO-ESP32S3-p-5627.html>

deepgram.com/pricing, checked Sep 22 2026

Deepgram pricing, Nova-3 streaming pay as you go

<https://deepgram.com/pricing>

Espressif ESP-IDF docs

ESP-IDF Wi-Fi Security guide (WPA2-Enterprise support on ESP32-S3)

<https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/api-guides/wifi-security.html>

